

OpenRouter + Chutes: A Minimal Contract-Game Framing

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1. What Is the Idea?

Current setup

- OpenRouter mostly buys pay-as-you-go API supply (S_A)
- This is flexible, but expensive for stable open-weight demand

Our proposal

- Add a wholesale quota layer (S_Q) from a provider such as Chutes
- Use token-normalized credits, not a naive request quota
- Keep OpenRouter's external API unchanged
- Let routing decide which requests deserve committed credits

Plain-language message

We are not just selling a better router.

We are proposing a new upstream procurement primitive:

spot API + wholesale quota

Our routing algorithm is what makes that quota economically valuable.

What the Chutes Data Tells Us

DeepSeek-V3.1 trace on Chutes

- 13.7M real requests over 30 days
- Request cost is highly skewed: $\text{Gini} = 0.54$
- A small fraction of expensive requests dominates the bill

Quota	Smart	Greedy	Gain
10%	38%	11%	3.6x
30%	67%	32%	2.1x
50%	83%	52%	1.6x

Interpretation

This is exactly the setting where wholesale quota can create value:

- stable enough demand to buy committed capacity
- enough cost skew that smart routing matters

Why this matters

Without smart routing, subscription procurement leaves a lot of value on the table.

2. Why Is This a Game-Theory Problem?

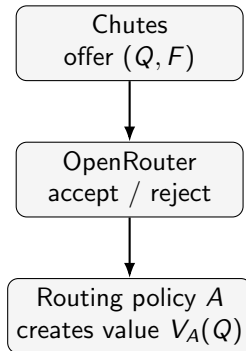
Why this is not just cost accounting

- Chutes chooses whether to offer a contract (Q, F)
- OpenRouter chooses whether to accept it
- If accepted, OpenRouter routes requests into the covered tranche

Minimal payoffs

$$\pi_{\text{OR}}(Q, F; A) = V_A(Q) - F$$

$$\pi_{\text{CH}}(Q, F; A) = F - \alpha_{\text{tranche}} V_A(Q) - \Delta_K$$



When does a win-win deal exist?

$$(1 - \alpha_{\text{tranche}}) V_A(Q) > \Delta_K$$

Interpretation: the new value created by the deal must exceed the provider's extra cost.

3. Why Does Our Algorithm Matter?

Core point

- The contract value is not fixed
- Better routing captures more of the possible savings:

$$V_A(Q) = \gamma_A(Q) V_{\text{opt}}(Q)$$

- So better routing expands the feasible contract space

Algorithm	Contract Zone	OR Savings
Greedy	\$12,676	16.7%
Best online (PD-EMA)	\$16,315	21.5%
Optimal	\$22,007	29.1%

What I want to confirm with the advisor

Should we present this as a bilateral contract game first, and treat auction as a later extension for the multi-provider setting?